

Implementation of SGPA and CGPA Calculation Using Nested Loops & Decision-Making Constructs in C

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1. Aim

To implement a C program for calculating Semester Grade Point Average (SGPA) for each semester and Cumulative Grade Point Average (CGPA) for multiple semesters using nested loops, input validation, and decision-making constructs.

2. Problem Statement

For m semesters, each semester contains n courses, where n may be different for different semesters. The program accepts a positive global ID, number of semesters, course code, credits, and grade. Valid grades are O, A, B, C, D, E, and P. It calculates SGPA for every semester and finally displays the CGPA.

3. Grade Point Conversion

O	A	B	C	D	E	P
10	9	8	7	6	5	4

4. Formulae Used

Semester Total Grade Credits (TGC) = $\sum (\text{Grade Point} \times \text{Course Credit})$

SGPA = Semester TGC / Semester Total Credits

CGPA = Cumulative TGC / Cumulative Total Credits

5. Algorithm

1. Start and declare the required integer, character, and floating-point variables.
2. Input `global_id` and repeat input until a positive value is entered.
3. Input the number of semesters m and repeat until m is positive.
4. Initialize cumulative total credits and cumulative total grade credits to zero.
5. For each semester, initialize semester totals to zero and input n , the number of courses. Validate n as positive.
6. For each course, input and validate a positive course code and positive credit value.
7. Input grade and accept only O, A, B, C, D, E, or P.
8. Convert the grade to grade point: O=10, A=9, B=8, C=7, D=6, E=5, P=4.
9. Add $GP \times \text{credit}$ to semester grade credits and add credit to semester total credits.
10. Calculate and display SGPA. Add semester totals to cumulative totals.

11. After all semesters, calculate CGPA and display global ID, number of semesters, cumulative credits, and CGPA.
12. Stop.

6. Flowchart / Program Logic

```
START
↓
Input & validate Global ID (> 0)
↓
Input & validate number of semesters m (> 0)
↓
Initialize cumulative totals = 0
↓
FOR each semester
↓
Input & validate n (> 0) → Reset semester totals
↓
FOR each course
↓
Input & validate code, credit and grade
↓
Convert grade → GP → Update semester totals
↓
END course loop
↓
Calculate & display SGPA → Update cumulative totals
↓
END semester loop
↓
Calculate & display CGPA
↓
STOP
```

7. C Program

```
#include <stdio.h>
#include <ctype.h>

int main() {
    int global_id, m, n, semester, course;
    int code, credit, GP;
    int semester_TC, semester_TGC;
    int cumulative_TC = 0, cumulative_TGC = 0;
    char grade;
    float SGPA, CGPA;

    do {
        printf("Enter Global ID: ");
        scanf("%d", &global_id);
        if (global_id <= 0)
            printf("Invalid Global ID. Enter a positive value.\n");
    } while (global_id <= 0);
```

```

do {
    printf("Enter number of semesters: ");
    scanf("%d", &m);
    if (m <= 0)
        printf("Invalid number of semesters. Enter a positive value.\n");
} while (m <= 0);

for (semester = 1; semester <= m; semester++) {
    semester_TC = 0;
    semester_TGC = 0;
    printf("\n--- Semester %d ---\n", semester);

    do {
        printf("Enter number of courses: ");
        scanf("%d", &n);
        if (n <= 0)
            printf("Invalid number of courses. Enter a positive value.\n");
    } while (n <= 0);

    for (course = 1; course <= n; course++) {
        printf("\nCourse %d\n", course);
        do {
            printf("Enter course code (positive integer): ");
            scanf("%d", &code);
            if (code <= 0)
                printf("Invalid course code.\n");
        } while (code <= 0);

        do {
            printf("Enter course credit: ");
            scanf("%d", &credit);
            if (credit <= 0)
                printf("Invalid credit. Enter a positive value.\n");
        } while (credit <= 0);

        do {
            printf("Enter grade (O/A/B/C/D/E/P): ");
            scanf("%c", &grade);
            grade = toupper((unsigned char)grade);
            if (grade!='O' && grade!='A' && grade!='B' && grade!='C' &&
                grade!='D' && grade!='E' && grade!='P')
                printf("Invalid grade. Please try again.\n");
        } while (grade!='O' && grade!='A' && grade!='B' && grade!='C' &&
            grade!='D' && grade!='E' && grade!='P');

        switch (grade) {
            case 'O': GP = 10; break;
            case 'A': GP = 9; break;
            case 'B': GP = 8; break;
            case 'C': GP = 7; break;
            case 'D': GP = 6; break;
            case 'E': GP = 5; break;
            default: GP = 4;
        }
        semester_TGC += GP * credit;
        semester_TC += credit;
    }
}

SGPA = (float)semester_TGC / semester_TC;

```

```

    printf("Semester %d SGPA = %.2f\n", semester, SGPA);
    cumulative_TGC += semester_TGC;
    cumulative_TC += semester_TC;
}

CGPA = (float)cumulative_TGC / cumulative_TC;
printf("\n===== FINAL RESULT =====\n");
printf("Global ID      : %d\n", global_id);
printf("No. of Semesters : %d\n", m);
printf("Cumulative Credits : %d\n", cumulative_TC);
printf("CGPA           : %.2f\n", CGPA);
return 0;
}

```

8. Test-Case Table

TC	Input / Condition	Type	Expected Result
1	global_id = 590032811, m = 1; 3 courses: (101,4,O), (102,3,A), (103,3,B)	Valid	SGPA = CGPA = 9.10; cumulative credits = 10
2	global_id = 0, then 590032811	Invalid → Valid	Error for 0; program asks again and accepts positive ID
3	m = 0, then 1	Invalid → Valid	Error for 0 semesters; program asks again
4	n = -2, then 1	Invalid → Valid	Error for negative course count; program asks again
5	course code = 0, then 101	Boundary/Invalid	0 rejected; positive code accepted
6	credit = 0, then 4	Boundary/Invalid	0 rejected; positive credit accepted
7	grade = X, then B	Invalid → Valid	X rejected; B accepted and GP = 8
8	m = 1, n = 1, code=101, credit=1, grade=P	Boundary Valid	Minimum positive counts accepted; SGPA = CGPA = 4.00
9	m = 1, n = 1, code=101, credit=1, grade=O	Boundary Valid	Highest grade accepted; SGPA = CGPA = 10.00

9. Sample Execution

```
Enter Global ID: 590032811
Enter number of semesters: 1

--- Semester 1 ---
Enter number of courses: 3

Course 1
Enter course code (positive integer): 101
Enter course credit: 4
Enter grade (O/A/B/C/D/E/P): O

Course 2
Enter course code (positive integer): 102
Enter course credit: 3
Enter grade (O/A/B/C/D/E/P): A

Course 3
Enter course code (positive integer): 103
Enter course credit: 3
Enter grade (O/A/B/C/D/E/P): B
Semester 1 SGPA = 9.10

===== FINAL RESULT =====
Global ID      : 590032811
No. of Semesters : 1
Cumulative Credits : 10
CGPA           : 9.10
```

10. Explanation of Important Constructs

Nested loops: The outer for-loop processes semesters and the inner for-loop processes all courses of the current semester.

do-while loops: Used for validation so that an input is requested at least once and repeated until it becomes valid.

switch statement: Maps each valid letter grade to its corresponding numerical grade point.

Type casting: (float) is used before division so SGPA and CGPA keep decimal values instead of integer division.

Cumulative totals: Credits and grade credits from every semester are added to obtain the final CGPA.

11. Result

The C program for SGPA and CGPA calculation was implemented using nested loops and decision-making constructs. It validates the required inputs, calculates semester-wise SGPA, and calculates the final CGPA using cumulative credits and grade credits. The prepared valid, invalid, and boundary test cases verify the required behavior.